

Lakshmi Bharathy Kumar

AI Engineer | LLM & Agentic Systems | Production ML Infrastructure

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Authorized to work in the U.S. without sponsorship

Summary

AI Engineer with an M.S. in Applied Data Intelligence from SJSU (May 2026) and 2 years of professional experience in data-driven marketing analytics. Focused on building production-grade LLM pipelines, agentic systems, and validation infrastructure, with emphasis on deterministic business logic, auditability, and cost-aware deployment over ad hoc prototyping.

Technical Skills

Languages: Python, SQL

LLMs & Agents: LangGraph, LangChain, Pydantic, Instructor, RAG, Prompt Engineering, Hugging Face, Groq / OpenAI APIs

Evaluation & Observability: Ragas, LangSmith, MLflow, Streamlit

ML & Deep Learning: PyTorch, ONNX, Quantization, scikit-learn, Pandas, NumPy

Infra & Tooling: FAISS, SQLite, Slack API, Docker, FastAPI, AWS, Git, CI/CD (GitHub Actions)

Projects

AI Customer Support Triage Agent

Python, LangGraph, FAISS, Groq, SQLite

- Built a production-oriented ticket triage system for a B2B SaaS environment with tiered SLA enforcement (Platinum/Diamond/Gold), decoupling tier-routing and SLA logic from LLM calls to keep business-critical rules deterministic and auditable.
- Designed a 30-article synthetic knowledge base with hierarchical markdown structure (sectioned content, embedded tables) to support parent-child RAG retrieval.
- Wired real-time Slack webhook alerts for escalated tickets, routing tier/priority/category context directly to the responsible team channel.

Autonomous Agentic Data Auditor

Python, Pydantic, Instructor, LLM Validation Pipelines

- Built an automated data ingestion and validation pipeline to parse unformatted, unstructured vendor text files at scale, enforcing strict schema and type-safety constraints on LLM outputs via Pydantic + Instructor to prevent malformed data from propagating downstream.
- Implemented a self-healing validation loop that detects structural anomalies and routes malformed payloads back for correction, capped at 3 retry cycles to control token cost.
- Raised input data consistency and schema completeness from 50% to 94%, delivering a reliable data layer for downstream relational databases.

Enterprise LLM Observability & Evaluation Engine

Ragas, LangSmith, MLflow, Python

- Built a monitoring dashboard to systematically evaluate LLM component performance, replacing manual ad hoc review.
- Automated evaluation pipelines using the Ragas framework to track Faithfulness, Answer Relevance, and Context Recall across releases.
- Instrumented telemetry to log token usage, request latency, and API errors per agent node.
- Cut average API operational costs 18% by identifying prompt redundancies and implementing targeted caching.

Image Restoration Model for Edge/NPU Deployment

PyTorch, ONNX, Quantization

- Designed a lightweight deep learning image enhancement model optimized for deployment on low-power edge processors.
- Exported PyTorch checkpoints to ONNX to standardize cross-platform hardware acceleration.
- Applied post-training quantization to reduce memory footprint and improve on-device inference efficiency.

Publications

L. B. Kumar, G. Liu, and Y. Luo, "Comparative Evaluation of Eye-Tracking Feature Representations for Gaze Direction Classification in Assistive Human-Machine Interaction," manuscript in preparation, 2026.

Education

M.S., Applied Data Intelligence

May 2026

San Jose State University, San Jose, CA

- Thesis** — Engineered 5 eye-tracking feature types (raw gaze, fixation, UFF, heatmaps, eyeball images) from Pupil Core sensor data; benchmarked 5 ML classifiers and 2 fine-tuned CNNs (ResNet18, MobileNetV2) — RF hit 0.90 user-dependent, SVM/GBDT reached 0.94 cross-user (LOUO), 12 pp above the published UFF+RF baseline.

B.E., Electronics and Communication Engineering

2014

Saranathan College of Engineering, Chennai, India

Experience

Digital Marketing Specialist

Jun 2021 – Dec 2022

RepuNEXT, Chennai, India

- Audited Google Analytics funnels to isolate drop-off points; updated page structure and SEO targeting based on findings, growing organic traffic 25%.
- Built automated reporting for weekly campaign metrics and audience segmentation; ROI improved 15%.